Visualization Literacy Analysis

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Read in data files

Raw data is in the `../AccData` subdirectory in two excel spreadsheet files per participant. The following code combines into single dataframe of all data.

```
#grab files from google drive (only have to do this once)
# source("getFromGoogleDrive.R")

#get the first 6? characters of each data file
#get unique values of these
# this is list of subject ids

raw_file_names <- list.files("../AccData")
first_six <- substr(raw_file_names, 1, 6)
sub_ids <- unique(first_six)


# fast_RTs <- data.frame(ParticipantId = character(),
#                        TrialName = character(),
#                        type = character(),
#                        time = numeric()
# )

all_data <- NULL

for (i in 1:length(sub_ids)){

  if (grepl("~$", sub_ids[i], fixed = T)){
    next
  }

  temp_file1 <- read_xlsx(paste0("../AccData/", sub_ids[i], "_1.xlsx")) %>%
    slice(1:17) %>%
    select(-starts_with("Order")) %>%
    rename(correct = 8)

  temp_file2 <- read_xlsx(paste0("../AccData/", sub_ids[i], "_2.xlsx")) %>%
    slice(1:17) %>%
     select(-starts_with("Order")) %>%
    rename(correct = 8)

  new_temp <- temp_file1 %>%
    bind_cols(temp_file2$correct) %>%
```

```r
    rename(Correct_1 = correct, Correct_2 = "...9") %>%
    mutate(AnswerRT = TimeToBeginInput - TimeToReadQuestion)


  all_data <- all_data %>%
    bind_rows(new_temp)

}

all_data <- all_data %>%
  rename(readRT = TimeToReadQuestion, totalRT = TimeToBeginInput)

#read in trialtype key (I created this from an early version of the previous paper)
trial_type_key <- read.csv("../trial_type_key.csv", stringsAsFactors = F)

all_data <- all_data %>%
  mutate(TrialType = trial_type_key$TrialType[match(TrialName, trial_type_key$TrialName)]) %>%
  mutate(TrialType = paste0("Type",TrialType))


# remove temporary dataframes to keep workspace cleaner as we go forward
rm(new_temp)
rm(temp_file1)
rm(temp_file2)
rm(trial_type_key)
```

Basic checks

Just a sanity check of all the data. Summary of number of subjects in each of the 3 condition types.

```r
#how many participants per condition
all_data %>%
  group_by(ParticipantId, Condition) %>%
  summarize(ntrials = n()) %>%
  group_by(Condition) %>%
  summarize(nsubs = n()) %>%
  summary()
```

```
## `summarise()` has grouped output by 'ParticipantId'. You can override using the `.groups`
## argument.
```

```
##   Condition             nsubs
##  Length:3           Min.   :33.00
##  Class :character   1st Qu.:36.00
##  Mode  :character   Median :39.00
##                     Mean   :40.67
##                     3rd Qu.:44.50
##                     Max.   :50.00
```

```r
#why is the balance so off?
```

Remove outliers based on RT

Removing outliers, resulting in data frame named `all_data_no_outliers`.

1. All trials where the `AnswerRT` is less than 2 seconds are removed first, which assumes these reactions were too fast so were done without an actual considered response.
2. All trials where the `AnswerRT` is 3 standard deviations different from the mean are removed next.

Table shows summary of mean and standard deviation of answer RT by each of the trial question types, as well as the 3 standard deviation upper and lower bound of the reaction time.

Displayed histogram is a sanity check of the number of trials each participant in all of the data have done. Most participants should do 16 trials. Less than that are from dropped data for outliers.

```
#removing on trial-by-trial basis


#remove answerRTs below 2000ms first
all_data_remove <- all_data %>%
  filter(AnswerRT >= 2000)
dim(all_data)[1]
```

```
## [1] 2074
```

```
dim(all_data_remove)[1]
```

```
## [1] 2067
```

```
#drops 7 trials

# also remove AnswerRT that are more than 3 standard deviations away from
# the mean
all_data_no_outliers <- all_data_remove %>%
  group_by(TrialName) %>%
  filter((!(abs(AnswerRT - mean(AnswerRT)) > 3.0*sd(AnswerRT))))
dim(all_data_no_outliers)[1]
```

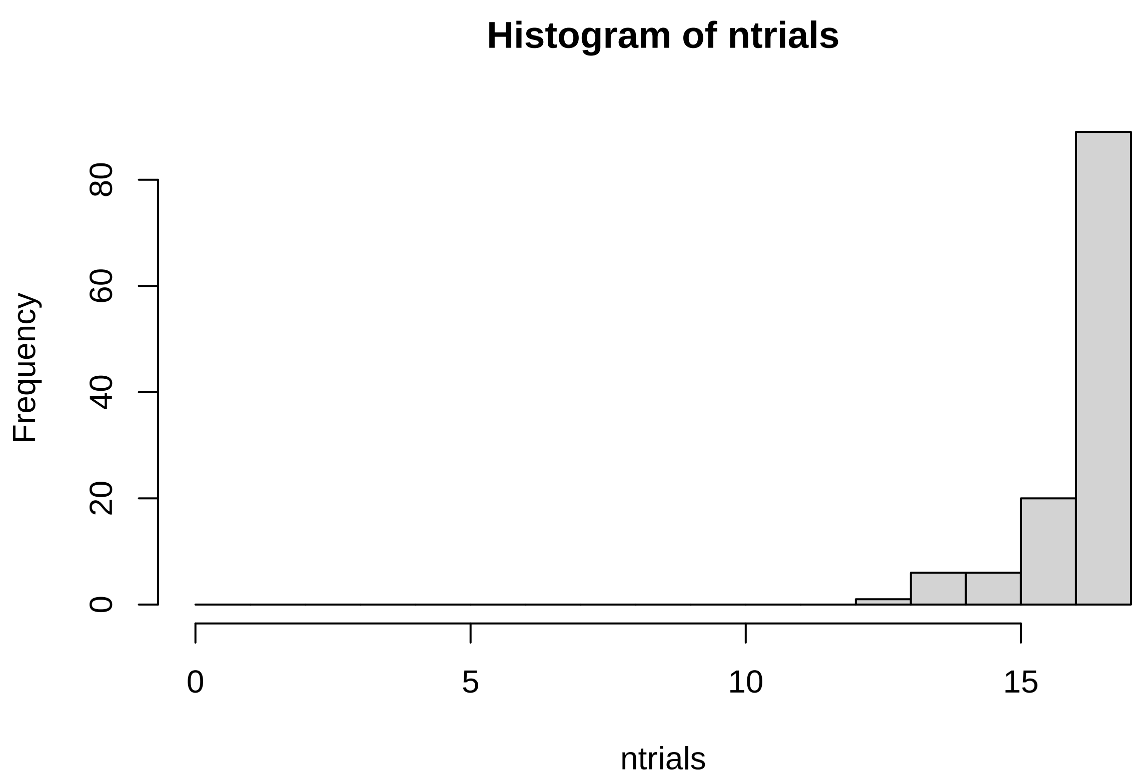
```
## [1] 2020
```

```
#drops 47 more trials


rt_data_summary <- all_data_no_outliers %>%
  group_by(TrialName) %>%
  summarize(meanAnswerRT = mean(AnswerRT, na.rm = T),
            sdAnswerRT = sd(AnswerRT, na.rm = T),
            LB = meanAnswerRT - 3*sdAnswerRT,
            UB = meanAnswerRT + 3*sdAnswerRT)
summary(rt_data_summary)
```

```
##   TrialName          meanAnswerRT     sdAnswerRT          LB                UB
##  Length:17          Min.   : 8963   Min.   : 5156   Min.   :-67692   Min.   : 24429
##  Class :character   1st Qu.:25275   1st Qu.:13263   1st Qu.:-35148   1st Qu.: 65715
##  Mode  :character   Median :29913   Median :19809   Median :-26012   Median : 88817
##                     Mean   :33477   Mean   :20886   Mean   :-29182   Mean   : 96137
##                     3rd Qu.:44519   3rd Qu.:26920   3rd Qu.:-17508   3rd Qu.:123649
##                     Max.   :61772   Max.   :43154   Max.   : -6504   Max.   :191235
```

```
rm(rt_data_summary)


all_data_no_outliers %>%
  group_by(ParticipantId) %>%
```

```
  summarize(ntrials = n()) %>%
  with(hist(ntrials, breaks = 0:17))
```




```
rm(all_data_remove)
##maybe consider replacing outliers with means instead of removing them?
```

Extract Only One Type of Chart

Create a dataframe from the data with no outliers that contains only XChartQ1 through XChartQ4 trials for the following analysis.

We replace the `all_data_no_outliers` dataframe with one containing only the selected TrialName question type here, so past this point we are analyzing only this particular TrialName type.

We show the table again of the mean RT for this question type. The table should match the one done with all of the data, but we should only have the particular question trial name type now in the data.

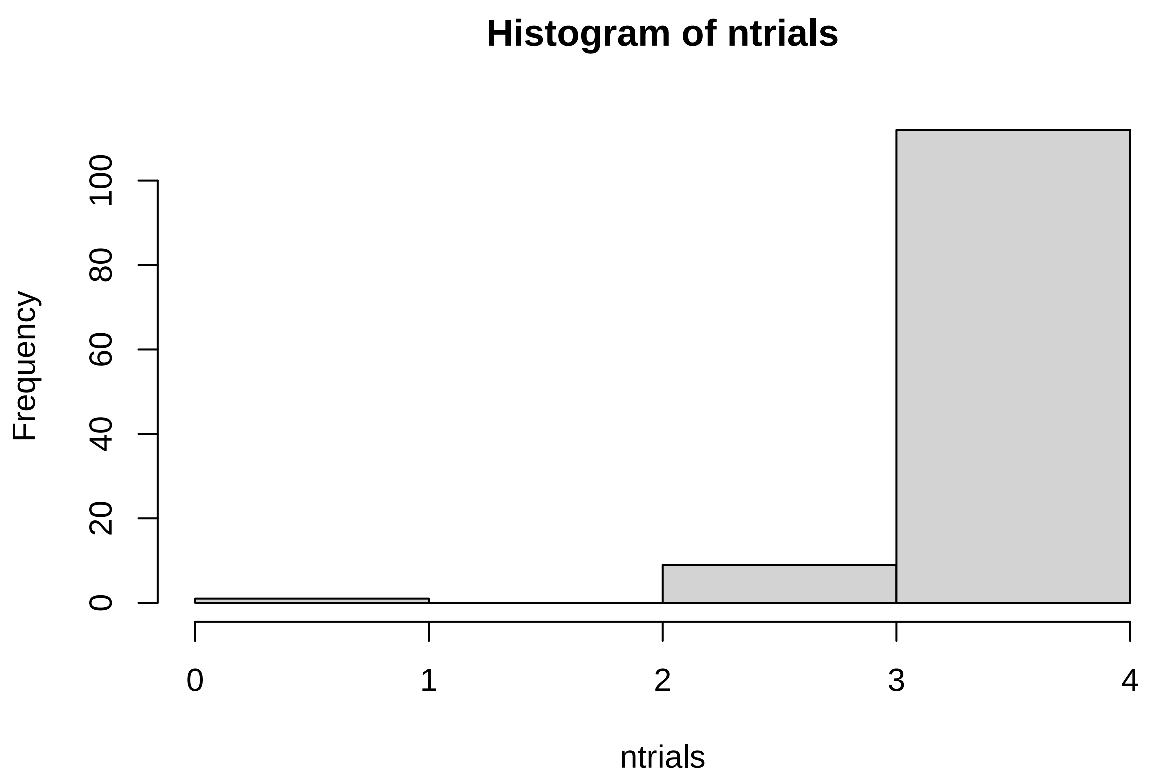
We also again plot the histogram of the number of trials. Participants should have seen 4 of each particular question type, but after removing outliers some only have 3, 2 or 1 remaining.

```
all_data_no_outliers <- all_data_no_outliers %>%
  filter(grepl("BarChartQ[0-9]", TrialName))

rt_data_summary <- all_data_no_outliers %>%
  group_by(TrialName) %>%
  summarize(meanAnswerRT = mean(AnswerRT, na.rm = T),
            sdAnswerRT = sd(AnswerRT, na.rm = T),
            LB = meanAnswerRT - 3*sdAnswerRT,
```

```r
            UB = meanAnswerRT + 3*sdAnswerRT)
summary(rt_data_summary)
```

```
##   TrialName          meanAnswerRT     sdAnswerRT          LB                UB
##  Length:4           Min.   : 8963   Min.   : 5156   Min.   :-17088   Min.   :24429
##  Class :character   1st Qu.:13267   1st Qu.: 7458   1st Qu.:-13507   1st Qu.:35641
##  Mode  :character   Median :15195   Median : 9575   Median :-11144   Median :43921
##                     Mean   :16513   Mean   : 9328   Mean   :-11470   Mean   :44497
##                     3rd Qu.:18441   3rd Qu.:11445   3rd Qu.: -9107   3rd Qu.:52777
##                     Max.   :26701   Max.   :13005   Max.   : -6504   Max.   :65715
```

```r
rm(rt_data_summary)


all_data_no_outliers %>%
  group_by(ParticipantId) %>%
  summarize(ntrials = n()) %>%
  with(hist(ntrials, breaks = 0:4))
```



Compare Answer RT

```r
#compare answer times

trial_type_means <- all_data_no_outliers %>%
  group_by(ParticipantId, Condition, TrialType) %>%
  summarize(mean_answerRT = mean(AnswerRT), n = n())
```

```
## `summarise()` has grouped output by 'ParticipantId', 'Condition'. You can override using
```

```
## the `.groups` argument.
```

```
# first, make .csv files in wide format to double check in statview
answer_rt <- trial_type_means %>%
  select(ParticipantId, Condition, TrialType, mean_answerRT) %>%
  pivot_wider(names_from = TrialType, values_from = mean_answerRT)

# fill in na values with mean for type1, type2 and type3 RT
# the following is supposed to work, but we get NaN from calculation of mean here, not sure why...
#read_rt_type_wider_clean <- read_rt_type_wider %>% mutate(across(Type1, ~replace_na(., mean(., na.rm=T
answer_rt <- answer_rt %>% replace_na(list(Type1 = mean(answer_rt$Type1, na.rm=TRUE)))
answer_rt <- answer_rt %>% replace_na(list(Type2 = mean(answer_rt$Type2, na.rm=TRUE)))
answer_rt <- answer_rt %>% replace_na(list(Type3 = mean(answer_rt$Type3, na.rm=TRUE)))


answer_rt_summary <- answer_rt %>%
  group_by(Condition) %>%
  summarize(meanType1 = mean(Type1, na.rm = T),
            meanType2 = mean(Type2, na.rm = T),
            meanType3 = mean(Type3, na.rm=T))
kable(answer_rt_summary)
```

Condition	meanType1	meanType2	meanType3
VR	22614.11	15243.81	11429.11
VR Monitor	29303.13	16692.65	13590.41
VR Monitor Stereo	29819.09	15174.25	11997.74

```
rm(answer_rt_summary)

#rm(trial_type_means)
```

Alternative Analysis using glmm TMB package

The following analysis is for the heterogenous group condition analysis. See .

The following determines if a more complex model is justified.

```
model_glmm_hom = glmmTMB(mean_answerRT ~ Condition * TrialType + (1 | ParticipantId),
                         data=trial_type_means, family = Gamma(link="log"),
                         dispformula = ~ 1, REML = FALSE)
model_glmm_hetero = glmmTMB(mean_answerRT ~ Condition * TrialType + (1 | ParticipantId),
                         data=trial_type_means, family = Gamma(link="log"),
                         dispformula = ~ Condition, REML = FALSE)


anova(model_glmm_hom, model_glmm_hetero)
```

```
## Data: trial_type_means
## Models:
## model_glmm_hom: mean_answerRT ~ Condition * TrialType + (1 | ParticipantId), zi=~0, disp=~1
## model_glmm_hetero: mean_answerRT ~ Condition * TrialType + (1 | ParticipantId), zi=~0, disp=~Conditi
##                   Df    AIC    BIC  logLik deviance  Chisq Chi Df Pr(>Chisq)
## model_glmm_hom    11 7517.9 7560.6 -3747.9   7495.9
## model_glmm_hetero 13 7521.4 7571.9 -3747.7   7495.4 0.4736      2     0.7892
```

Try overall effect reporting

```r
fixed_effects_anova <- Anova(model_glmm_hom, type="III")
kable(fixed_effects_anova)
```

	Chisq	Df	Pr(>Chisq)
(Intercept)	16539.823090	1	0.0000000
Condition	6.164242	2	0.0458619
TrialType	61.158375	2	0.0000000
Condition:TrialType	4.027610	4	0.4022821

```r
#summary(fixed_effects_anova, level=0.95)
```

EMM on Condition and Interaction

If the glmm analysis is not significant, just use the homogeneous model. Otherwise report on the more complex heterogeneous model?

```r
emm_main_condition <- emmeans(model_glmm_hom,
                              specs = ~ Condition,
                              type = "response")
```

```
## NOTE: Results may be misleading due to involvement in interactions
```

```r
condition_results <- pairs(emm_main_condition)


summary(condition_results, level=0.95)
```

```
##  contrast                       ratio     SE  df null z.ratio p.value
##  VR / VR Monitor                0.857 0.0740 Inf    1  -1.782  0.1755
##  VR / VR Monitor Stereo         0.902 0.0817 Inf    1  -1.137  0.4909
##  VR Monitor / VR Monitor Stereo 1.052 0.1000 Inf    1   0.532  0.8555
##
## Results are averaged over the levels of: TrialType
## P value adjustment: tukey method for comparing a family of 3 estimates
## Tests are performed on the log scale
```

```r
confint(condition_results, level=0.95)
```

```
##  contrast                       ratio     SE  df asymp.LCL asymp.UCL
##  VR / VR Monitor                0.857 0.0740 Inf     0.700      1.05
##  VR / VR Monitor Stereo         0.902 0.0817 Inf     0.729      1.12
##  VR Monitor / VR Monitor Stereo 1.052 0.1000 Inf     0.841      1.32
##
## Results are averaged over the levels of: TrialType
## Confidence level used: 0.95
## Conf-level adjustment: tukey method for comparing a family of 3 estimates
## Intervals are back-transformed from the log scale
```

```r
#kable(condition_results)
```

```r
emm_interaction <- emmeans(model_glmm_hom,
                           specs = ~ Condition * TrialType,
                           type = "response")
```

```r
simple_effects_results <- pairs(emm_interaction, by = "TrialType")


summary(simple_effects_results, level=0.95)
```

```
## TrialType = Type1:
##  contrast                      ratio     SE  df null z.ratio p.value
##  VR / VR Monitor               0.789 0.0921 Inf    1  -2.026  0.1061
##  VR / VR Monitor Stereo        0.768 0.0938 Inf    1  -2.161  0.0780
##  VR Monitor / VR Monitor Stereo 0.973 0.1250 Inf    1  -0.216  0.9746
##
## TrialType = Type2:
##  contrast                      ratio     SE  df null z.ratio p.value
##  VR / VR Monitor               0.917 0.1080 Inf    1  -0.741  0.7390
##  VR / VR Monitor Stereo        0.984 0.1230 Inf    1  -0.133  0.9902
##  VR Monitor / VR Monitor Stereo 1.073 0.1390 Inf    1   0.542  0.8508
##
## TrialType = Type3:
##  contrast                      ratio     SE  df null z.ratio p.value
##  VR / VR Monitor               0.871 0.1010 Inf    1  -1.189  0.4600
##  VR / VR Monitor Stereo        0.972 0.1180 Inf    1  -0.235  0.9700
##  VR Monitor / VR Monitor Stereo 1.116 0.1430 Inf    1   0.853  0.6700
##
## P value adjustment: tukey method for comparing a family of 3 estimates
## Tests are performed on the log scale
```

```r
confint(simple_effects_results, level=0.95)
```

```
## TrialType = Type1:
##  contrast                      ratio     SE  df asymp.LCL asymp.UCL
##  VR / VR Monitor               0.789 0.0921 Inf     0.601      1.04
##  VR / VR Monitor Stereo        0.768 0.0938 Inf     0.577      1.02
##  VR Monitor / VR Monitor Stereo 0.973 0.1250 Inf     0.720      1.31
##
## TrialType = Type2:
##  contrast                      ratio     SE  df asymp.LCL asymp.UCL
##  VR / VR Monitor               0.917 0.1080 Inf     0.696      1.21
##  VR / VR Monitor Stereo        0.984 0.1230 Inf     0.734      1.32
##  VR Monitor / VR Monitor Stereo 1.073 0.1390 Inf     0.791      1.45
##
## TrialType = Type3:
##  contrast                      ratio     SE  df asymp.LCL asymp.UCL
##  VR / VR Monitor               0.871 0.1010 Inf     0.664      1.14
##  VR / VR Monitor Stereo        0.972 0.1180 Inf     0.731      1.29
##  VR Monitor / VR Monitor Stereo 1.116 0.1430 Inf     0.826      1.51
##
## Confidence level used: 0.95
## Conf-level adjustment: tukey method for comparing a family of 3 estimates
## Intervals are back-transformed from the log scale
```

```r
#kable(simple_effects_results)
```

EMM by VR, VR Monitor, VR Monitor Stereo

```
emm_main_condition <- emmeans(model_glmm_hom,
                              specs = ~ Condition,
                              type = "response")
```

```
## NOTE: Results may be misleading due to involvement in interactions
```

```
summary(emm_main_condition, level=0.95)
```

```
##  Condition          response   SE  df asymp.LCL asymp.UCL
##  VR                    15184  871 Inf     13568     16992
##  VR Monitor            17708 1140 Inf     15602     20098
##  VR Monitor Stereo     16832 1180 Inf     14667     19316
##
## Results are averaged over the levels of: TrialType
## Confidence level used: 0.95
## Intervals are back-transformed from the log scale
```

```
confint(emm_main_condition, level=0.95)
```

```
##  Condition          response   SE  df asymp.LCL asymp.UCL
##  VR                    15184  871 Inf     13568     16992
##  VR Monitor            17708 1140 Inf     15602     20098
##  VR Monitor Stereo     16832 1180 Inf     14667     19316
##
## Results are averaged over the levels of: TrialType
## Confidence level used: 0.95
## Intervals are back-transformed from the log scale
```

Try explicitly naming the coefficients to remove all ambuugity

```
comparison_results <- contrast(
  emm_main_condition,
  method = list(
    VR_vs_Monitors_Collapsed = c('VR' = 1, 'VR Monitor' = -0.5, 'VR Monitor Stereo' = -0.5)
  )
)

summary(comparison_results, infer = TRUE)
```

```
##  contrast                 ratio     SE  df asymp.LCL asymp.UCL null z.ratio p.value
##  VR_vs_Monitors_Collapsed 0.879 0.0656 Inf      0.76      1.02    1  -1.723  0.0849
##
## Results are averaged over the levels of: TrialType
## Confidence level used: 0.95
## Intervals are back-transformed from the log scale
## Tests are performed on the log scale
```

```
confint(comparison_results, level = 0.95)
```

```
##  contrast                 ratio     SE  df asymp.LCL asymp.UCL
##  VR_vs_Monitors_Collapsed 0.879 0.0656 Inf      0.76      1.02
##
## Results are averaged over the levels of: TrialType
## Confidence level used: 0.95
## Intervals are back-transformed from the log scale
```

Try explicitly naming the coefficients to remove all ambuugity

```
comparison_results <- contrast(
  emm_main_condition,
  method = list(
    VRMonitor_vs_others_collapsed = c('VR' = -0.5, 'VR Monitor' = 1, 'VR Monitor Stereo' = -0.5)
  )
)

summary(comparison_results, infer = TRUE)
```

```
##  contrast                      ratio     SE  df asymp.LCL asymp.UCL null z.ratio p.value
##  VRMonitor_vs_others_collapsed  1.11 0.0873 Inf     0.949      1.29    1   1.297  0.1945
##
## Results are averaged over the levels of: TrialType
## Confidence level used: 0.95
## Intervals are back-transformed from the log scale
## Tests are performed on the log scale
```

```
confint(comparison_results, level = 0.95)
```

```
##  contrast                      ratio     SE  df asymp.LCL asymp.UCL
##  VRMonitor_vs_others_collapsed  1.11 0.0873 Inf     0.949      1.29
##
## Results are averaged over the levels of: TrialType
## Confidence level used: 0.95
## Intervals are back-transformed from the log scale
```

Compare Accuracy

Extract wide dataframe

Do all of the previous again but on the Accuracy results.

We again create a dataframe in needed format of only the accuracy data.
A bit different from before, only need to create an avverage of the two correct features we have. But then we do a similar transformation into wide format and replace na/missing values.

```
all_data_no_outliers <- all_data_no_outliers %>%
  rowwise() %>%
  mutate(Correct_Avg = mean(c(Correct_1, Correct_2)))

trial_type_mean_accuracy <- all_data_no_outliers %>%
  group_by(ParticipantId, Condition, TrialType) %>%
  summarize(mean_acc = mean(Correct_Avg),
            n = n())
```

```
## `summarise()` has grouped output by 'ParticipantId', 'Condition'. You can override using
## the `.groups` argument.
```

```
accuracy <- trial_type_mean_accuracy %>%
  select(ParticipantId, Condition, TrialType, mean_acc) %>%
  pivot_wider(names_from = TrialType, values_from = mean_acc)

# replace na with mean
accuracy <- accuracy %>% replace_na(list(Type1 = mean(accuracy$Type1, na.rm=TRUE)))
accuracy <- accuracy %>% replace_na(list(Type2 = mean(accuracy$Type2, na.rm=TRUE)))
accuracy <- accuracy %>% replace_na(list(Type3 = mean(accuracy$Type3, na.rm=TRUE)))
```

```
#rm(trial_type_mean_accuracy)
```

Apply transformation of mean accuracy to shift so no longer have exactly 0 and 1 values.

```
num_trials <- nrow(all_data_no_outliers)
trial_type_mean_accuracy$transformed_acc <- (trial_type_mean_accuracy$mean_acc * (trial_type_mean_accur
```

Alternative Analysis using glmm TMB package

Use all data no outliers for alternative Bernoulli/Binomial GLMM

```
all_data_no_outliers$transformed_acc <- (all_data_no_outliers$Correct_Avg > 0.74) * 1.0
```

```
model_glmm_hom = glmmTMB(transformed_acc ~ Condition * TrialType + (1 | ParticipantId),
                            data=all_data_no_outliers, family = binomial(link="logit"),
                            REML = FALSE)
#model_glmm_hetero = glmmTMB(transformed_acc ~ Condition * TrialType + (1 | ParticipantId),
#                             data=all_data_no_outliers, family = binomial(link="logit"),
#                             dispformula = ~ Condition, REML = FALSE)


#anova(model_glmm_hom, model_glmm_hetero)
```

```
fixed_effects_anova <- Anova(model_glmm_hom, type="III")
kable(fixed_effects_anova)
```

	Chisq	Df	Pr(>Chisq)
(Intercept)	24.0100707	1	0.0000010
Condition	0.0207802	2	0.9896637
TrialType	0.8187910	2	0.6640516
Condition:TrialType	0.8226226	4	0.9353897

```
emm_main_condition <- emmeans(model_glmm_hom,
                          specs = ~ Condition,
                          type = "response")
```

```
## NOTE: Results may be misleading due to involvement in interactions
```

```
condition_results <- pairs(emm_main_condition)


summary(condition_results, level=0.95)
```

```
##  contrast                        odds.ratio      SE  df null z.ratio p.value
##  VR / VR Monitor                   1.27e+00 1.88000 Inf    1   0.159  0.9862
##  VR / VR Monitor Stereo            2.00e-07 0.00245 Inf    1  -0.001  1.0000
##  VR Monitor / VR Monitor Stereo    1.00e-07 0.00193 Inf    1  -0.001  1.0000
##
## Results are averaged over the levels of: TrialType
## P value adjustment: tukey method for comparing a family of 3 estimates
## Tests are performed on the log odds ratio scale
```

```
confint(condition_results, level=0.95)
```

```
##  contrast                        odds.ratio      SE  df asymp.LCL asymp.UCL
```

```
##  VR / VR Monitor                  1.27e+00 1.88000 Inf     0.039        41
##  VR / VR Monitor Stereo           2.00e-07 0.00245 Inf     0.000       Inf
##  VR Monitor / VR Monitor Stereo   1.00e-07 0.00193 Inf     0.000       Inf
##
## Results are averaged over the levels of: TrialType
## Confidence level used: 0.95
## Conf-level adjustment: tukey method for comparing a family of 3 estimates
## Intervals are back-transformed from the log odds ratio scale
```

```r
#kable(condition_results)
```

```r
emm_interaction <- emmeans(model_glmm_hom,
                           specs = ~ Condition * TrialType,
                           type = "response")

simple_effects_results <- pairs(emm_interaction, by = "TrialType")


summary(simple_effects_results, level=0.95)
```

```
## TrialType = Type1:
##  contrast                       odds.ratio       SE  df null z.ratio p.value
##  VR / VR Monitor                     0.874 1.670000 Inf    1  -0.071  0.9972
##  VR / VR Monitor Stereo              1.257 3.070000 Inf    1   0.094  0.9952
##  VR Monitor / VR Monitor Stereo      1.439 3.640000 Inf    1   0.144  0.9887
##
## TrialType = Type2:
##  contrast                       odds.ratio       SE  df null z.ratio p.value
##  VR / VR Monitor                     3.148 5.870000 Inf    1   0.615  0.8121
##  VR / VR Monitor Stereo              0.000 0.000002 Inf    1  -0.001  1.0000
##  VR Monitor / VR Monitor Stereo      0.000 0.000001 Inf    1  -0.001  1.0000
##
## TrialType = Type3:
##  contrast                       odds.ratio       SE  df null z.ratio p.value
##  VR / VR Monitor                     0.738 1.160000 Inf    1  -0.194  0.9795
##  VR / VR Monitor Stereo              0.000 0.000001 Inf    1  -0.002  1.0000
##  VR Monitor / VR Monitor Stereo      0.000 0.000001 Inf    1  -0.002  1.0000
##
## P value adjustment: tukey method for comparing a family of 3 estimates
## Tests are performed on the log odds ratio scale
```

```r
confint(simple_effects_results, level=0.95)
```

```
## TrialType = Type1:
##  contrast                       odds.ratio       SE  df asymp.LCL asymp.UCL
##  VR / VR Monitor                     0.874 1.670000 Inf   0.01000        76
##  VR / VR Monitor Stereo              1.257 3.070000 Inf   0.00411       384
##  VR Monitor / VR Monitor Stereo      1.439 3.640000 Inf   0.00382       541
##
## TrialType = Type2:
##  contrast                       odds.ratio       SE  df asymp.LCL asymp.UCL
##  VR / VR Monitor                     3.148 5.870000 Inf   0.03972       249
##  VR / VR Monitor Stereo              0.000 0.000002 Inf   0.00000       Inf
##  VR Monitor / VR Monitor Stereo      0.000 0.000001 Inf   0.00000       Inf
##
## TrialType = Type3:
```

```
##  contrast                        odds.ratio       SE  df asymp.LCL asymp.UCL
##  VR / VR Monitor                      0.738 1.160000 Inf   0.01883        29
##  VR / VR Monitor Stereo               0.000 0.000001 Inf   0.00000       Inf
##  VR Monitor / VR Monitor Stereo       0.000 0.000001 Inf   0.00000       Inf
##
## Confidence level used: 0.95
## Conf-level adjustment: tukey method for comparing a family of 3 estimates
## Intervals are back-transformed from the log odds ratio scale
```

```r
#kable(simple_effects_results)
```

EMM by VR, VR Monitor, VR Monitor Stereo

```r
emm_main_condition <- emmeans(model_glmm_hom,
                              specs = ~ Condition,
                              type = "response")
```

```
## NOTE: Results may be misleading due to involvement in interactions
```

```r
summary(emm_main_condition, level=0.95)
```

```
##  Condition              prob       SE  df asymp.LCL asymp.UCL
##  VR                0.9997584 3.67e-04 Inf 0.9952790 0.9999877
##  VR Monitor        0.9996942 4.75e-04 Inf 0.9936119 0.9999854
##  VR Monitor Stereo 1.0000000 5.91e-07 Inf 0.0000000 1.0000000
##
## Results are averaged over the levels of: TrialType
## Confidence level used: 0.95
## Intervals are back-transformed from the logit scale
```

```r
confint(emm_main_condition, level=0.95)
```

```
##  Condition              prob       SE  df asymp.LCL asymp.UCL
##  VR                0.9997584 3.67e-04 Inf 0.9952790 0.9999877
##  VR Monitor        0.9996942 4.75e-04 Inf 0.9936119 0.9999854
##  VR Monitor Stereo 1.0000000 5.91e-07 Inf 0.0000000 1.0000000
##
## Results are averaged over the levels of: TrialType
## Confidence level used: 0.95
## Intervals are back-transformed from the logit scale
```